

EX. 1. Find an IP over $\mathbb{F}^{m \times n}$ $(A, B) = \text{tr}(B^T A)$

2. Let V, W be IPs. Find an IP over $V \times W$, V^*

Furthermore, if N is a subspace of V , V/N

3. Find IP over $\mathbb{R}[x]$

$$2. \mathcal{Q}((v, w), (v', w')) = (v, v') + (w, w')$$

$$\textcircled{2} \dim V = n < \infty \quad v_1, \dots, v_n$$

$$V^* \text{ dual basis } v_1^* \dots v_n^* \longrightarrow (v_i^*, v_j^*) = (v_i, v_j)$$

Application Riesz Representation Thm (豪斯表示定理)

Let V be an f.d. IPS / $\mathbb{R}$. Then there is a natural isomorphism between V and its dual V^*

$$\text{pf. } \sigma: V \longrightarrow V^*$$

$$v \longmapsto \sigma(v) = u \longrightarrow \sigma(v)(u) = (v, u)$$

It's enough to show that σ is inj ($\dim V = \dim V^*$)

$$v \in \ker \sigma \quad \text{i.e. } \sigma(v) = 0, \text{ so } \forall u \in V, (v, u) = 0 \Rightarrow v = 0$$

Multiplication of vector spaces

Question $V, W/\mathbb{F}$ def T s.t. $\dim T = \dim U \times \dim V$

EX $\dim \text{Hom}(V, W) = \dim \mathbb{F}^{\dim W \times \dim V} = \dim W \times \dim V$

Try to define $V \otimes W = \text{Hom}(V, W)$ (not bad, but not natural)

Physics. $V \otimes W = \text{span} \{ |v_i, w_j\rangle \}$

ENG Quotient (Orthogonal projection)

$$V = M \oplus M^\perp \xrightarrow{\text{quotient}} M \cong V/M^\perp$$

Step 1. Construct the free space $F(V \times W)$

EX. Let X be any set. The free space on X is

$$F(X) = \text{span } X = \{ v \in F(X) \mid v = \sum_{i=1}^n a_i x_i, a_i \in \mathbb{F}, x_i \in X \}$$

Step 2. $T = V \otimes W = F(V \times W) / N$. What's N ?

$$N = \text{span} \left\{ \begin{array}{l} (v, w) + (v', w) - (v+v', w) \\ (v, w) + (v, w') - (v, w+w') \\ \lambda(v, w) - (\lambda v, w), \lambda(v, w) - (v, \lambda w) \end{array} \right\}$$

$$\text{so in } F(V \times W) / N, \lambda[(v, w) + N] = (\lambda v, w) + N = (v, \lambda w) + N$$

$$(v, w) + N = v \otimes w \quad \lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$$

$$\text{EX. } \mathbb{R}[x, y] = \mathbb{R}[x] \otimes \mathbb{R}[y]$$

$$\text{EX. } \mathbb{R} \times \mathbb{R} = \mathbb{R}^2, \mathbb{R} \otimes \mathbb{R} = F(\mathbb{R} \times \mathbb{R}) / N, N =$$

$$\text{Conclusion, } \dim \mathbb{R} \otimes \mathbb{R} = 1$$

• Universal Construction (泛构造, 万能构造)

Step 1. Bilinear map

$$V \times W \xrightarrow{t} T$$

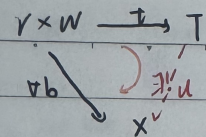
$$(v, w) \longmapsto t(v, w)$$

$$(v+v', w) \longmapsto t(v+v', w) = t(v, w) + t(v', w)$$

$$(v, w+w') \longmapsto t(v, w+w') = t(v, w) + t(v, w')$$

$$(\lambda v, w) \longmapsto t(\lambda v, w) = \lambda t(v, w)$$

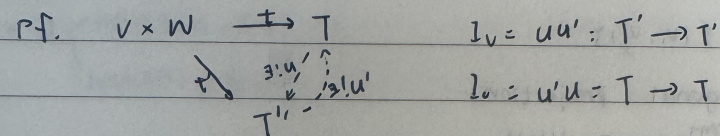
$$(v, \lambda w) \longmapsto t(v, \lambda w) = \lambda t(v, w)$$



By Bil. t, $\exists! u \in \text{Hom}(T, X)$ s.t. $b = ut$

Then (T, t) is the tensor product of V and W
 $V \otimes W$

Thm Tensor Product is unique up to isomorphism



$$I_V = u'u': T' \rightarrow T'$$

$$I_W = u'u': T' \rightarrow T'$$